Artist identification with a multi-view DINOv2 fusion model (V3.2)

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Problem statement

We continue the artist identification study on digitised paintings. V1 established an 18-artist baseline, V2 added unknown rejection and V3 trained on 39 known artists. V3.2 keeps the 39-artist task and tests three combined changes: a larger DINOv2 encoder, multi-view inputs that expose fine brushwork, and a multi-prototype unknown-artist gate. The model reaches 84.81% known accuracy on a held-out test split, but the combined change, a new split and a single seed mean this is not a controlled improvement over V3, and the unknown-artist gate exceeded its false-acceptance target.

Datasets and measured results

Version and scope	Retained images	Known test	Accuracy	Macro F1
V1 / 18 artists	4,743	726	85.54%	85.90%
V2 / 50 known artists	8,418 incl. unknowns	1,158	63.47%	61.87%
V3 / 39 known artists	15,824 incl. unknowns	1,945	77.02%	76.20%
V3.2 / 39 known artists (new split)	16,087 incl. unknowns	1,949	84.81%	83.93%

V3.2 uses the same WikiArt source and 39 known artists as V3, but a fresh seeded random split that follows the V3 per-artist counts, so its test paintings are not the V3 test paintings. Rows are different recognition tasks and splits, not a common leaderboard. Totals include development and held-out unknown cohorts.

Approach

V3.2 builds on the V3 pipeline: DINOv2 embeddings fused with 58 HSV/LBP and 324 HOG values, a logistic warm start for the classifier head, conservative fine-tuning and confidence/prototype rejection. Three components change. (1) The encoder is DINOv2-base (768 values) in place of the small model (384). (2) Each painting yields five 224-pixel views: one aspect-preserving global view and a 2 x 2 grid of local tiles cut from a 448-pixel copy; the five embeddings are averaged. (3) Each artist is represented by three k-means prototypes instead of one, and the gate mixes calibrated confidence with the best prototype similarity. The fused feature has 768 + 58 + 324 = 1,150 values. These changes were run together, so their individual effects are not separated.

Implementation

A Python package implements manifest-driven data loading, multi-view preprocessing, handcrafted descriptors, the fusion model, resumable training, calibrated rejection, evaluation and unit tests. Image shards are verified against SHA-256 checksums before extraction, and the manifest is built from the audited per-image quality file. Training ran locally on an RTX 4060 laptop GPU (8 GB) with bf16 mixed precision, gradient checkpointing and per-epoch checkpoints. The configuration and per-epoch history are saved with each run.

V3.2 training and evaluation

V3.2 retained 16,087 of 16,088 candidates. Known splits contain 7,727 training, 1,925 validation, 1,286 calibration and 1,949 test paintings. Unknown development, calibration and test contain 800, 800 and 1,600 images from separate artist cohorts.

Fine-tuning updates the final two encoder blocks and normalisation with learning rate 0.000002, and the classifier with 0.0001. Dropout 0.25, weight decay 0.03, label smoothing 0.05, square-root inverse-frequency class weights, mild augmentation and early stopping constrain fitting. The effective batch is 32 (4 x 8 accumulation). Validation selected epoch 38; training stopped at epoch 46 of a maximum 60, in 2.69 hours (210.6 s per epoch). The confidence temperature (0.75), the confidence/prototype mix (0.6) and the threshold (0.885) were fitted on validation, calibration and unknown-development data; the test split was evaluated only after these were fixed.

V3.2 measure	Observed result (test split)
Known accuracy / macro F1 / top-5	84.81% / 83.93% / 96.15%
Balanced accuracy / calibration error (ECE) / NLL	84.05% / 0.0202 / 0.559
Known paintings accepted	989 / 1,949 = 50.74%
Correct among accepted known paintings	982 / 989 = 99.29%
Unknown paintings falsely accepted	130 / 1,600 = 8.13% (target 5%)
Known versus unknown AUROC	0.8023
Weakest artists (test F1)	Polenov 53.3%, Surikov 59.5%, Levitan 66.7%, Delacroix 67.9%
Strongest artists (test F1)	Bouguereau 97.7%, Russell 96.5%, Escher 96.2%, Yerka 95.7%

Conclusions

Test accuracy is 84.81% and macro F1 is 83.93%, against 77.02% and 76.20% for V3. This should not be read as a measured improvement: the split differs from V3, the encoder, views and gate changed together, no V3 baseline was retrained on this split, and there is one training seed. The diagnostic training macro F1 at the selected epoch is 100%, compared with 83.37% validation F1, a gap of 16.6 points (V3: 22.9 points), so a generalisation concern remains.

The unknown-artist gate accepted 50.74% of known test paintings with 99.29% accuracy among them, but it falsely accepted 8.13% of unknown test paintings. The threshold gave exactly 5.00% on the unknown calibration cohort, so the 5% target did not transfer to unseen artists. V3 operated at a different point (24.47% accepted, 1.89% false acceptance), so the two gates cannot be ranked, and the AUROC of 0.8023 against 0.7585 compares different splits. The 99.29% conditional score is not overall recognition, and unknown rejection is not artwork authentication. Remaining issues include imperfect duplicate screening, possible foundation-model pretraining overlap, source bias and the weak Russian realist and landscape artists.

Proposed extension

The next step is to retrain the V3 setup (small encoder, global view) on this exact split, then add the base encoder, the multi-view input and the multi-prototype gate one at a time, with seeds 107 and 108 for a mean and spread. A second backbone, CutMix-style regularisation and a more conservative false-acceptance target are candidates to test. A fresh confirmation set is needed because the V3.2 test split has now been inspected. No performance gain is claimed for any component before this evaluation.

Sources: Chen and Deng (2018), DINOv2 (Oquab et al.), dataset records and archived run files (history.json, results_dev.json, results_final.json). Source code: github.com/gurupreetham16/artist-identification.